AI-Based Student Performance Prediction and Early Warning System

By

Achal Yadav
T055 TYIT

studying in

Sheth L.U.J and Sir M.V College, Andheri

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Artificial Intelligence and Application Development

2. Introduction:

- Student performance is important for colleges.
- Academic data is available.
- Manual identification of weak students takes time.
- Machine learning can identify patterns.
- The proposed system predicts student performance.

3. Problem Statement

To develop an AI-based system that analyzes student academic and behavioral information and predicts whether a student is likely to pass or fail, while identifying students who require early academic intervention.

4. Objectives

- To collect student academic data.
- To preprocess the dataset.
- To identify important performance-related features.
- To train a machine-learning classification model.
- To predict student performance.

- To identify high-risk students.
- To provide early recommendations.
- To evaluate model performance.

5. Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

6. Dataset

- **Dataset name -** Student Performance
- **Source of dataset -**

  https://www.kaggle.com/dsv/13489035?utm_source=chatgpt.com
- **Number of records -** 500 records
- **Important attributes/features -** Attendance ,Study_Hours

,Assignment_Score ,Internal_Marks ,Previous_CGPA ,Backlogs ,Extracurricular ,Final_Result

• **Target variable -** Final_Result

7. Methodology / Working

- **Data Collection** – Load the student performance dataset containing academic and behavioral information.
- **Data Preprocessing** – Clean the data and convert categorical values into numerical values.
- **Feature Selection** – Select important features such as attendance, study hours, internal marks, previous CGPA, and backlogs.
- **Data Splitting** – Divide the dataset into training and testing data.
- **Model Training** – Train a **Random Forest Classifier** using the training data.
- **Prediction** – Predict whether a student will **PASS or FAIL**.
- **Evaluation** – Calculate the model's accuracy using the test dataset.
- **Result Analysis** – Use the predictions to identify students who

may require additional academic support.

8. Algorithm Used

Random Forest Classifier – A supervised machine learning algorithm that combines multiple decision trees to predict the student's Final_Result (PASS/FAIL) accurately.

9. Implementation

```python
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# Load student data
df = pd.read_csv("student_performance_500.csv")

# Convert categorical data
le = LabelEncoder()

df["Gender"] = le.fit_transform(df["Gender"])
df["Extracurricular"] = le.fit_transform(df["Extracurricular"])
df["Final_Result"] = le.fit_transform(df["Final_Result"])

# Select features
features = [
    "Age", "Gender", "Attendance", "Study_Hours",
    "Assignment_Score", "Internal_Marks",
    "Previous_CGPA", "Backlogs", "Extracurricular"
]

X = df[features]
y = df["Final_Result"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train Random Forest model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Student Performance Prediction")
print("------------------------------")
print("Model Accuracy:", round(accuracy * 100, 2), "%")
```

The Python program loads the student performance dataset from a CSV file, preprocesses the categorical data, selects relevant academic

features, and trains a Random Forest classification model. The trained model predicts student performance and evaluates its accuracy using the test dataset.

10. Results

```
Enter Attendance (%): 80
Enter Study Hours per Day: 4
Enter Assignment Score: 5
Enter Internal Marks: 25
Enter Previous CGPA: 8.7
Enter Number of Backlogs: 0
```

```
Student Performance Prediction
------------------------------
Model Accuracy: 93.0 %
Prediction: PASS
[[0.33 0.67]]
Risk Level: MEDIUM
```

11. Advantages

- Early identification of weak students
- Reduces manual analysis
- Helps faculty intervention

12. Limitations

- Prediction depends on dataset quality.
- Small datasets may not generalize well.
- Prediction is not a guarantee of actual academic performance.
- Student behavior can change over time.

13. Applications

- Colleges
- Schools
- Universities
- Coaching institutes
- Online learning platforms

14. Conclusion

The project successfully predicts student performance using Random Forest. It helps identify students at risk and supports early academic intervention.

15. Future Scope

- Add a web interface.
- Connect the model to a database.
- Automatically send alerts to faculty.
- Use historical data to track student performance over multiple semesters.

16. References

- ChatGPT: https://chatgpt.com/

- Google: https://google.com/